

Deep Generative Models: Variational Inference and Expectation Maximization

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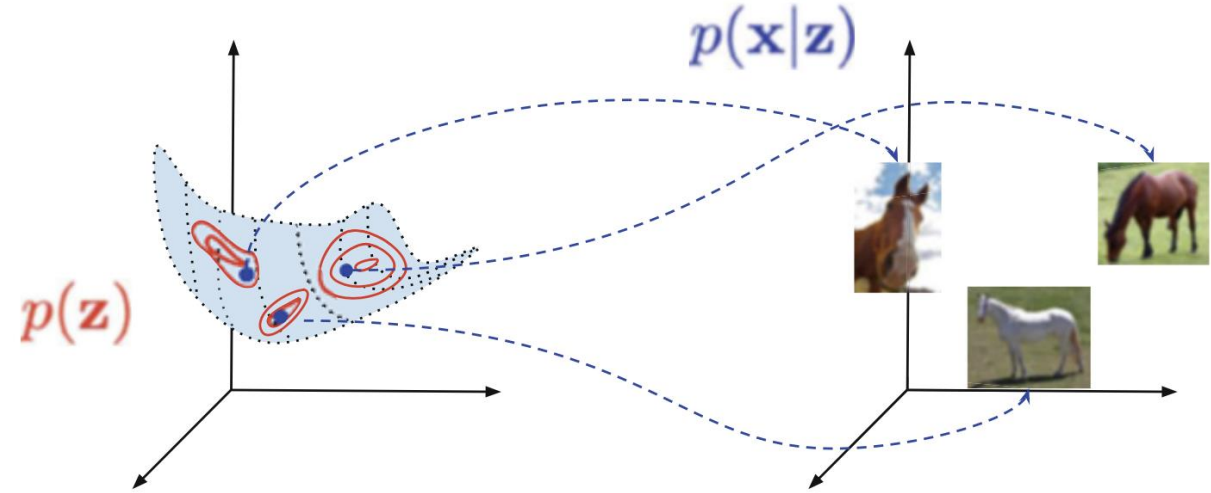


Outline

- **Latent Variable Models**
 - Probabilistic PCA
 - Beyond PPCA?
- Variational Inference
 - Principle
 - Derivation
- Expectation Maximization
 - Derivation
 - EM for a Mixture of Gaussians

Latent Variable Models

- X = observed variable
- Z = latent variable
- $\mathbf{z} \sim p(\mathbf{z})$
- $\mathbf{x} \sim p(\mathbf{x} | \mathbf{z})$



A latent variable model and a generative process. Note the low-dimensional manifold (here 2D) embedded in the high-dimensional space (here 3D)

- Factorization of the joint model

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x} | \mathbf{z})p(\mathbf{z})$$

- Marginalization of the model

$$p(\mathbf{x}) = \int p(\mathbf{x} | \mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Probabilistic Principal Component Analysis (PPCA)

- Let $\mathbf{z} \sim \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon} \mid \mathbf{0}, \sigma^2 \mathbf{I})$ be independent.
- In PPCA, the dependency between $\mathbf{x} \in \mathbb{R}^D$ and $\mathbf{z} \in \mathbb{R}^d$, $d \ll D$, is defined by a linear Gaussian additive model

$$\mathbf{x} = \mathbf{W}\mathbf{z} + \mathbf{b} + \boldsymbol{\epsilon}$$

- **Theorem.** Let $\boldsymbol{\mu}_N$ and $\boldsymbol{\Sigma}_N$ be, respectively, the ML estimates for the mean and the covariance of the data. Let $\mathbf{U}_1$ be the matrix with the top d eigenvectors of $\boldsymbol{\Sigma}_N$, $\boldsymbol{\Lambda}_1$ be the matrix with the corresponding top d eigenvalues, and λ_i be the i th largest eigenvalue of $\boldsymbol{\Sigma}_N$. The ML estimates for the PPCA parameters $(\mathbf{b}, \mathbf{W}, \sigma)$ are given by

$$\mathbf{b} = \boldsymbol{\mu}_N, \mathbf{W} = \mathbf{U}_1(\boldsymbol{\Lambda}_1 - \sigma^2 \mathbf{I})^{1/2} \mathbf{R} \text{ and } \sigma^2 = \frac{1}{D - d} \sum_{i=d+1}^D \lambda_i$$

where $\mathbf{R} \in \mathbb{R}^{d \times d}$ is an arbitrary orthogonal matrix.

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What about Latent Variable Models other than PPCA

- We would like to learn the parameters of the model via Maximum Likelihood.
- Since $\mathbf{z}$ is latent, we need to marginalize $p_{\theta}(\mathbf{x}) = \int p_{\theta}(\mathbf{x} \mid \mathbf{z})p(\mathbf{z})d\mathbf{z}$, i.e.

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i) = \max_{\theta} \sum_{i=1}^N \log \int p_{\theta}(\mathbf{x}_i \mid \mathbf{z})p(\mathbf{z})d\mathbf{z}$$

- For PPCA, we could compute $p(\mathbf{x})$ in closed form and solve for θ analytically.
- In general, we need many samples of $\mathbf{z}$ for each $\mathbf{x}_i$ to approximate the integral.

$$\max_{\theta} \sum_{i=1}^N \log \sum_{j=1}^N p_{\theta}(\mathbf{x}_i \mid \mathbf{z}_j)$$

- We address this challenge using **Variational Inference**, which we describe next.

Variational Inference

- **Old ML learning objective:**

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(x_i) = \max_{\theta} \sum_{i=1}^N \log \int p_{\theta}(x_i | z) p(z) dz$$

- **Theorem:** the log likelihood can be written as

$$\log p_{\theta}(\mathbf{x}) = \max_{q(\cdot|\mathbf{x}): q(\cdot|\mathbf{x}) \geq 0, \int q(z|\mathbf{x}) dz = 1} \int q(z | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, z)}{q(z|\mathbf{x})} dz.$$

and the maximizing distribution is given by $q^*(\mathbf{z} | \mathbf{x}) = p_{\theta}(\mathbf{z} | \mathbf{x})$

- **New ML learning objective:**

$$\max_{\theta} \max_{q(\cdot|\mathbf{x}_i), \forall i} \sum_{i=1}^N \int q(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z} | \mathbf{x}_i)} dz$$

- Before going through the derivation, what is the gain here?

Variational Inference

- **New ML learning objective:**

$$\max_{\theta} \max_{q(\cdot | \mathbf{x}_i), \forall i} \sum_{i=1}^N \int q(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z} | \mathbf{x}_i)} d\mathbf{z}$$

- **Expectation Maximization:**

- If finding q^* given θ is easy, then we can alternate between finding q given θ and vice versa
- Here $q^*(\mathbf{z} | \mathbf{x}) = p_{\theta}(\mathbf{z} | \mathbf{x})$. Thus, if the integral w.r.t. $\mathbf{z}$ is easy to evaluate for a fixed θ , we can alternate between computing the integral (E-step) and maximizing w.r.t. θ (M-step).

- **Variational AutoEncoders:** represent $q(\cdot | x_i)$ with a function with parameters ψ (e.g., a DNN) that maps x_i to a distribution $q_{\psi}(\cdot | x_i)$, and find (θ, ψ) via SGD

$$\max_{\theta} \max_{\psi} \sum_{i=1}^N \int q_{\psi}(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q_{\psi}(\mathbf{z} | \mathbf{x}_i)} d\mathbf{z}$$

Variational Inference

- **New ML learning objective:**

$$\max_{\theta} \max_{q(\cdot|\mathbf{x}_i), \forall i} \sum_{i=1}^N \int q(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z} | \mathbf{x}_i)} d\mathbf{z}$$

- We will use VI for many latent variable models
 - Mixtures of Gaussians (a.k.a. Gaussian Mixture Models) → EM
 - Probabilistic Principal Component Analysis (PPCA) → EM
 - Mixtures of PPCA → EM
 - Variational Auto-Encoders (VAE) → VI
 - Diffusion models → VI
 - ...

Variational Inference: Derivation

- Proof: Let $q(z|x)$ be the variational distribution. Observe that

$$\begin{aligned}\log p_{\theta}(x) &= \int q(\mathbf{z} | \mathbf{x}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} = \int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z} | \mathbf{x})} d\mathbf{z} \\ &= \int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} | \mathbf{x})} \frac{q(\mathbf{z} | \mathbf{x})}{p_{\theta}(\mathbf{z} | \mathbf{x})} d\mathbf{z} \\ &= \boxed{\int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} | \mathbf{x})} d\mathbf{z}} + \boxed{\int q(\mathbf{z} | \mathbf{x}) \log \frac{q(\mathbf{z} | \mathbf{x})}{p_{\theta}(\mathbf{z} | \mathbf{x})} d\mathbf{z}} \\ &\quad \text{Evidence Lower Bound (ELBO)} \quad \text{KL}[q(\mathbf{z} | \mathbf{x}) || p_{\theta}(\mathbf{z} | \mathbf{x})]\end{aligned}$$

- Thus, $\max_{q(\cdot|x)} \int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} | \mathbf{x})} d\mathbf{z} = \max_{q(\cdot|x)} \log p_{\theta}(\mathbf{x}) - \text{KL}(q(\mathbf{z} | \mathbf{x}) || p_{\theta}(\mathbf{z} | \mathbf{x}))$
 $= \log p_{\theta}(\mathbf{x}) - \min_{q(\cdot|x)} \text{KL}(q(\mathbf{z} | \mathbf{x}) || p_{\theta}(\mathbf{z} | \mathbf{x})) = \log p_{\theta}(\mathbf{x})$
- The last step follows because $\text{KL}(q || p) \geq 0$ and $\text{KL}(q || p) = 0$ iff $p = q$.

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Expectation Maximization

- ML objective

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i) \equiv \max_{\theta} \max_{q(\cdot|\mathbf{x}_i), \forall i} \sum_{i=1}^N \int q(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z} | \mathbf{x}_i)} d\mathbf{z}$$

- Expectation Maximization alternates between two steps (k : iteration)

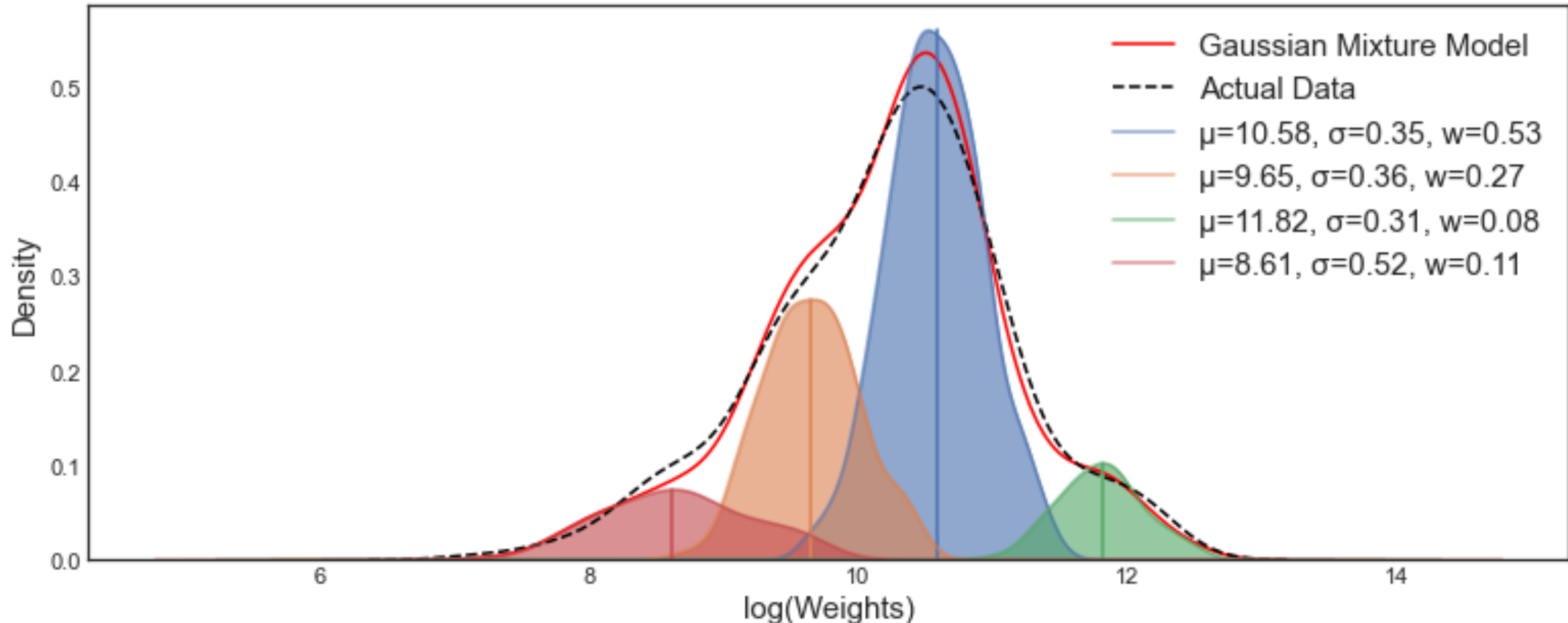
- **E-step:** $q^k(\mathbf{z} | \mathbf{x}_i) = p_{\theta^k}(\mathbf{z} | \mathbf{x}_i)$ maximize w.r.t. q with θ fixed
- **E-step:** $Q(\theta | \theta_k) = \sum_{i=1}^N \int_{\mathbf{z}} q^k(\mathbf{z} | \mathbf{x}_i) \log p_{\theta}(\mathbf{x}_i, \mathbf{z}) d\mathbf{z}$ integrate w.r.t. q with θ fixed
- **M-step:** $\theta^{k+1} = \operatorname{argmax}_{\theta} Q(\theta | \theta^k)$ maximize w.r.t. θ with q fixed

- Examples

- For PPCA and for a mixture of Gaussians, E & M steps are closed-form (HW1, next slide).
- E-step often done by sampling (MCMC) and M-step often done by optimization (SGD).
- The E-step is the hardest: it requires an “easy to sample from” distribution.

Example: EM for Gaussian Mixture Model (GMM)

- **GMM:** $p_{\theta}(\mathbf{x}) = \sum_{j=1}^n p(\mathbf{x} \mid z = j)p(z = j) = \pi_1 p_{\theta_1}(\mathbf{x}) + \cdots + \pi_n p_{\theta_n}(\mathbf{x})$
 - $\pi_j := p(z = j) > 0$: prior probability of drawing a point from the j -th model; $\sum_{j=1}^n \pi_j = 1$
 - $p_{\theta_j}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu_j, \Sigma_j)$, $\theta_j = (\mu_j, \Sigma_j)$ are the mean and covariance of the j -th Gaussian
 - $\theta = (\theta_1, \dots, \theta_n, \pi_1, \dots, \pi_n)$ are the parameters of the mixture model



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 - $\theta = (\theta_1, \dots, \theta_n, \pi_1, \dots, \pi_n)$ are the parameters of the mixture model
- **Goal:** estimate θ from N i.i.d. samples $x_1, \dots, x_N$ from p_{θ} using EM
 - For $i = 1, \dots, N$, let $z_i = j$ if x_i belongs to class j , and let $q_{ij} = q(z_i = j \mid \mathbf{x}_i)$

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i) = \max_{\theta} \max_{q_{ij} \geq 0: \sum_j q_{ij} = 1, \forall i} \sum_{i=1}^N \sum_{j=1}^n q_{ij} \log \frac{p_{\theta}(\mathbf{x}_i, z_i = j)}{q_{ij}}$$

- **E-step:** compute $q_{ij}^k = p_{\theta^k}(z_i = j \mid \mathbf{x}_i) = \frac{p_{\theta^k}(\mathbf{x}_i \mid z_i = j)p_{\theta^k}(z_i = j)}{p_{\theta^k}(\mathbf{x}_i)} = \boxed{\frac{p_{\theta_j^k}(\mathbf{x}_i)\pi_j^k}{\sum_{l=1}^n p_{\theta_l^k}(\mathbf{x}_i)\pi_l^k}}$
- **E-step:** the expectation w.r.t z is easy to compute because z is discrete.

Example: EM for Gaussian Mixture Model (GMM)

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 - $\pi_j := p(z = j) > 0$: prior probability of drawing a point from the j -th model; $\sum_{j=1}^n \pi_j = 1$
 - $p_{\theta_j}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu_j, \Sigma_j)$, $\theta_j = (\mu_j, \Sigma_j)$ are the mean and covariance of the j -th Gaussian
 - $\theta = (\theta_1, \dots, \theta_n, \pi_1, \dots, \pi_n)$ are the parameters of the mixture model

- **M-step:**

$$\theta^{k+1} = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log \frac{p_{\theta}(\mathbf{x}_i, z_i=j)}{q_{ij}^k} = \underset{\{\theta_j\}_{j=1}^n, \{\pi_j\}_{j=1}^n}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log(\pi_j \cdot p_{\theta_j}(\mathbf{x}_i))$$

- π_j 's: $\max_{\{\pi_j\}_{j=1}^n: \sum_{j=1}^n \pi_j = 1} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log(\pi_j)$. The solution is given by $\pi_j^{k+1} = \frac{\sum_{i=1}^N q_{ij}^k}{\sum_{i=1}^N \sum_{j=1}^n q_{ij}^k}$

- θ_j 's: $\max_{\mu_j, \Sigma_j} \sum_{i=1}^N q_{ij}^k \left(-\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_j)^{\top} \Sigma_j^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_j) - \frac{1}{2} \log \det(\Sigma_j) \right)$. The solution is

$$\boldsymbol{\mu}_j^{k+1} = \frac{\sum_{i=1}^N q_{ij}^k \mathbf{x}_i}{\sum_{i=1}^N q_{ij}^k} \quad \text{and} \quad \Sigma_j^{k+1} = \frac{\sum_{i=1}^N q_{ij}^k (\mathbf{x}_i - \boldsymbol{\mu}_j^{k+1})(\mathbf{x}_i - \boldsymbol{\mu}_j^{k+1})^{\top}}{\sum_{i=1}^N q_{ij}^k}$$

E.g.: EM for Gaussian Mixture Model

- **M-step:** $\theta^{k+1} = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log \frac{p_{\theta}(x_i, z_i=j)}{q_{ij}^k} = \underset{\{\theta_j\}_{j=1}^n, \{\pi_j\}_{j=1}^n}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log(\pi_j \cdot p_{\theta_j}(x_i))$

- For π_j 's: $\max_{\{\pi_j\}_{j=1}^n: \sum_{j=1}^n \pi_j = 1} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log(\pi_j)$

- To analyze the solution of a constrained optimization problem, we use the method of Lagrange multipliers:

$$L(\{\pi_j\}_j, \lambda) = \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log(\pi_j) - \lambda(\sum_{j=1}^n \pi_j - 1)$$

- $\frac{\partial L}{\partial \lambda} = 0 \Leftrightarrow \sum_{j=1}^n \pi_j = 1$

- $\forall j: \frac{\partial L}{\partial \pi_j} = 0 \Leftrightarrow \sum_{i=1}^N q_{ij}^k \frac{1}{\pi_j} - \lambda = 0$

- The solution is $\lambda = \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k, \pi_j = \frac{\sum_{i=1}^N q_{ij}^k}{\sum_{i=1}^N \sum_{j=1}^n q_{ij}^k} =: \pi_j^{k+1}$

E.g.: EM for Gaussian Mixture Model

- **M-step:** $\theta^{k+1} = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log \frac{p_{\theta}(x_i, z_i=j)}{q_{ij}^k} = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log p_{\theta}(x_i, z_i=j)$
 - If we spell out θ and $p_{\theta}(x_i, z_i=j)$, the above becomes $\underset{\{\theta_j\}_{j=1}^n, \{\pi_j\}_{j=1}^n}{\operatorname{argmax}} \sum_{i=1}^N \sum_{j=1}^n q_{ij}^k \log (\pi_j \cdot p_{\theta_j}(x_i))$
- For θ_j 's: $\max_{\mu_j, \Sigma_j} \left[\sum_{i=1}^N q_{ij}^k \left(-\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_j)^{\top} \Sigma_j^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_j) - \frac{1}{2} \log \det(\Sigma_j) \right) \right] =: \mathcal{L}(\boldsymbol{\mu}_j, \Sigma_j)$
- $\frac{\partial \mathcal{L}}{\partial \boldsymbol{\mu}_j} = \sum_{i=1}^N q_{ij}^k \Sigma_j^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_j) = \Sigma_j^{-1} \left(\sum_{i=1}^N q_{ij}^k \mathbf{x}_i - \left(\sum_{i=1}^N q_{ij}^k \right) \boldsymbol{\mu}_j \right).$
- Setting it to 0 $\Rightarrow \boldsymbol{\mu}_j = \frac{\sum_{i=1}^N q_{ij}^k \mathbf{x}_i}{\sum_{i=1}^N q_{ij}^k} =: \boldsymbol{\mu}_j^{k+1}$
- $\mathcal{L}(\boldsymbol{\mu}_j, \Sigma_j) = -\frac{1}{2} \operatorname{tr} \left(\Sigma_j^{-1} \sum_{i=1}^N q_{ij}^k (\mathbf{x}_i - \boldsymbol{\mu}_j) (\mathbf{x}_i - \boldsymbol{\mu}_j)^{\top} \right) - \frac{1}{2} \left(\sum_{i=1}^N q_{ij}^k \right) \log \det \Sigma_j$
- Reusing our derivation in MLE for Gaussian: $\Sigma_j = \frac{\sum_{i=1}^N q_{ij}^k (\mathbf{x}_i - \boldsymbol{\mu}_j^{k+1}) (\mathbf{x}_i - \boldsymbol{\mu}_j^{k+1})^{\top}}{\sum_{i=1}^N q_{ij}^k} =: \Sigma_j^{k+1}$